

GRADUATE READING

Architecting Big Data & AI Infrastructure

*Principles, trade-offs, and the
Mercedes-AMG PETRONAS F1
case*

Graduate-level overview for Data Analytics & AI
courses

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1. Why this topic matters

For most of the last decade, "Big Data" and "Artificial Intelligence" have been discussed largely as analytical capabilities. The conversation has now shifted decisively toward what an organization must build to do those things at scale.

Training a modern foundation model requires tens of thousands of GPUs running for months; serving it requires globally distributed inference clusters; and feeding it requires data pipelines that ingest, clean, and version petabytes of heterogeneous information. The question facing decision-makers is no longer "should we adopt AI?" but "what kind of computational, data, and operational infrastructure will let us adopt it sustainably?"

This article lays out the architectural principles that a graduate student or practitioner should weigh when designing such infrastructure, and then grounds them in a working case — the Mercedes-AMG PETRONAS Formula 1 Team — where the abstract principles become very concrete decisions about cloud platforms, edge devices, and AI workloads.

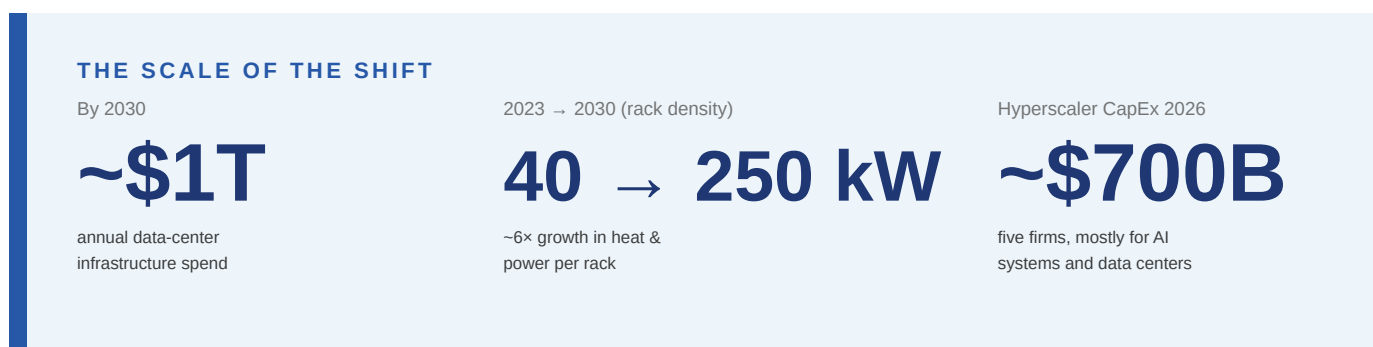


Figure 1. *The scale of the AI-driven infrastructure build-out. Source: IoT Analytics (2025); Data Center Knowledge (2026); industry estimates.*

2. The shape of the problem

Before discussing principles, it helps to picture the workload. A modern data-and-AI platform must usually do all of the following at once:

- **Ingest** data continuously from many heterogeneous sources (sensors, logs, transactions, third-party feeds, documents, images).
- **Store** that data in formats and tiers appropriate to its value and access pattern — hot, warm, cold; row-based, column-based, vector, graph.
- **Process** it in two very different modes: batch (overnight, periodic) and streaming (real-time or near-real-time).
- **Train** models on historical data, which is compute-intensive but tolerant of latency.
- **Serve** model predictions, which is latency-sensitive but usually individually small.
- **Govern** all of the above for security, privacy, lineage, reproducibility, cost, and regulatory compliance.

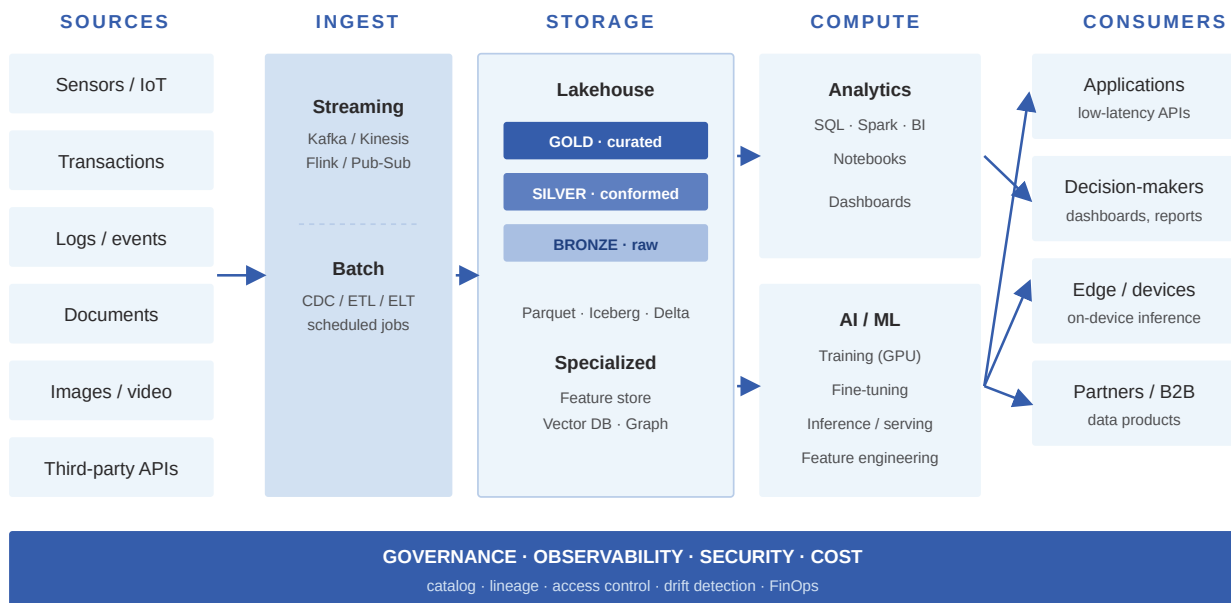


Figure 2. A reference architecture for a modern data + AI platform. Governance and observability are not a sidebar — they cut across every layer.

A useful mental model is the well-known "three Vs" of Big Data — Volume, Velocity, Variety — extended to a "five Vs" framework that adds **Veracity** (data quality and trustworthiness) and **Value** (relevance to a decision). Each V has direct architectural consequences.

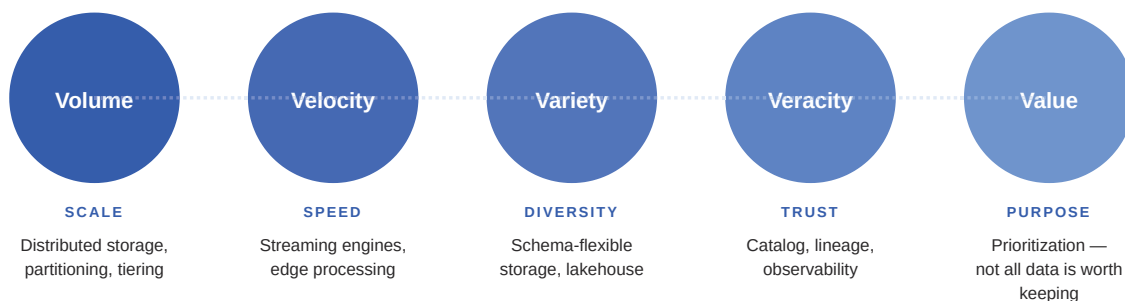


Figure 3. The five V's of Big Data. Each "V" pushes the architecture in a different direction — and good design respects all five at once.

3. Principles for defining the architecture

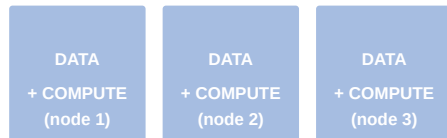
The following principles draw from established practice in data engineering (Kleppmann, 2017; Reis & Housley, 2022) and machine-learning systems (Huyen, 2022). They are deliberately neutral about specific vendors, because the underlying decisions repeat regardless of whether one builds on AWS, Azure, Google Cloud, or a hybrid stack.

01

Separate storage from compute

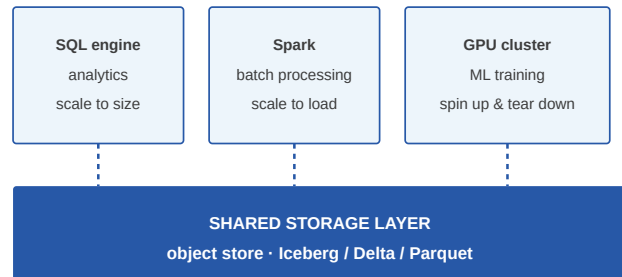
Historically, systems like the original Hadoop tied data to the machines that processed it. Modern architectures decouple the two: data sits in object storage or a lakehouse layer, and elastic compute clusters attach to it on demand. This lets organizations scale either dimension independently, retire idle compute without losing data, and allow multiple specialized engines to read the same tables. For AI workloads — which are bursty and expensive — this is non-negotiable.

MONOLITHIC (LEGACY)



Scaling means buying both, together.
Idle compute is impossible to retire
without moving the data.

DECOUPLED (MODERN)



Same data, multiple engines. Each scales independently.

Figure 4. Storage / compute separation. Multiple specialized engines attach on demand to a shared data layer.

02

Design for the data lifecycle, not just the data

Data has a lifecycle — raw, conformed, curated, served — and good architectures make this explicit. The contemporary expression of this idea is the **medallion architecture** (bronze, silver, gold layers) or, more generally, the **lakehouse**: a single platform that combines the schema flexibility and cost profile of a data lake with the transactional guarantees and performance of a warehouse. One team's "raw landing zone" becomes another team's "trusted source," without copying data across systems.



Each layer adds quality, trust, and aggregation — without ever copying data out of the platform.

Figure 5. The medallion architecture: bronze → silver → gold. The lakehouse pattern makes the data lifecycle explicit.

03

Treat batch and streaming as one continuum

Lambda architectures, which maintained parallel batch and streaming pipelines, are increasingly replaced by **unified processing** models (Apache Beam, Spark Structured Streaming, Flink) where the same logic can run over either bounded or unbounded data. The architectural principle is: do not duplicate business logic across engines. If a metric is defined twice, it will eventually be defined two different ways, and the inconsistency will be discovered at the worst possible moment.

04

Push computation to where the data is — and to where the decision is

Centralization gives you economies of scale, but latency-critical decisions must be made close to the user, the sensor, or the device. Modern architectures include **edge** components (on-premises clusters, in-vehicle compute, in-factory gateways) tightly integrated with a central cloud. The choice is not "edge or cloud" but "what runs where, and how do they synchronize?"

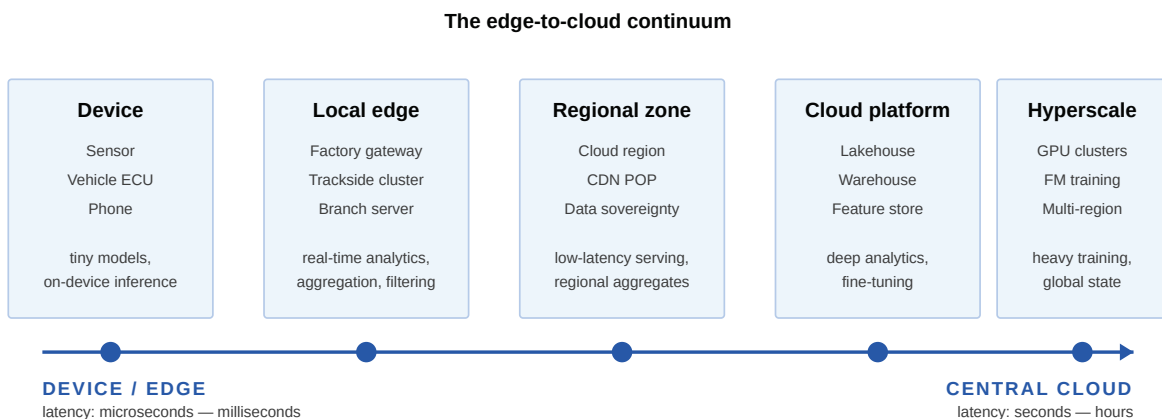


Figure 6. *The edge-to-cloud continuum. The architectural question is not "edge or cloud" but "what runs where?"*

05

Build for elasticity, but plan for constraint

Cloud elasticity gives the illusion of infinite capacity. The reality is that GPU supply, power availability, and data-center construction are now hard constraints on AI growth — hyperscalers are reportedly investing hundreds of billions of dollars in 2025–2026 specifically to relieve these bottlenecks. Good architectures are designed to **degrade gracefully** under capacity pressure: workloads are tiered by criticality, non-essential training can be paused, and inference can fall back to smaller models if needed.

06

Make the platform observable end-to-end

A Big Data + AI platform is not "done" when it works. It is done when, when it fails, you can tell *why*. This requires unified observability across data quality, pipeline health, model behavior, and infrastructure. Without it, debugging a production AI system becomes an act of archaeology.

07

Govern by design, not by audit

Privacy, lineage, access control, and reproducibility cannot be retrofitted. The architectural principle is to encode governance into the platform itself: every dataset has an owner; every pipeline produces lineage metadata; every model has a versioned training run; every access is logged. This is doubly important in regulated industries but pays off everywhere by shortening the time to answer "where did this number come from?"

08

Optimize for total cost, not just unit cost

A GPU-hour is cheap; a training run that has to be restarted from scratch because someone deleted a dataset is not. The relevant cost is the **end-to-end cost of producing a trusted result**, including engineering time, debugging time, and the opportunity cost of slow iteration.

09

Modularity and standard interfaces

A platform that locks you into a specific vendor, framework, or file format will eventually be expensive to change. Open formats (Parquet, Iceberg, ONNX), standard APIs (SQL, REST, OpenAI-compatible inference endpoints), and modular service boundaries preserve optionality. This matters most precisely when the field is evolving quickly, which it is.

10

Sustainability as a first-class constraint

Power consumption is no longer an externality. Rack densities have climbed from roughly 40 kW per rack in 2023 to over 100 kW in 2025, with 250 kW projected by 2030. Cooling, water use, and grid impact are now part of any honest infrastructure conversation. Designing for energy efficiency — through workload scheduling, model compression, and locating compute where clean power is available — is becoming a competitive and ethical requirement.

Centralized vs. Edge	Elasticity vs. Predictability	Best-of-breed vs. Integrated
CENTRAL FAVORS - Economies of scale - Easier governance - Deep analytics EDGE FAVORS - Lower latency - Resilience offline - Data sovereignty	ELASTIC FAVORS - Pay-as-you-go - Scale on demand - No idle capacity RESERVED FAVORS - Lower unit cost - Deterministic perf. - Capacity guaranteed	BEST-OF-BREED FAVORS - Best tool per job - Fastest innovation - Avoid lock-in INTEGRATED FAVORS - Fewer integrations - Simpler ops - Lower TCO

Figure 7. Three trade-offs every architect must weigh. No universal answer — only context-dependent ones.

4. The benefits of integrating Artificial Intelligence

Why integrate AI into a Big Data platform at all, rather than treating them as separate concerns? Several benefits emerge specifically from the integration.

4.1 What AI brings to data

Automated feature extraction. Traditional analytics requires humans to specify what to measure. ML models, particularly deep ones, learn relevant features directly from raw data — meaning the platform can extract value from sources (images, audio, free text, telemetry streams) that would be impractical to feature-engineer manually.

Adaptive systems. A reporting system tells you what happened. An AI system can adjust its predictions as conditions change — drift in customer behavior, seasonality, new regulations — provided the infrastructure supports continuous retraining and deployment.

Decision support at machine speed. Many decisions in modern industry (fraud scoring, ad bidding, network routing, race strategy) must be made faster than any human can intervene. AI integrated into the data pipeline closes this loop.

Better use of unstructured data. Foundation models — LLMs and their multimodal cousins — have made the 80% of enterprise data that is unstructured (emails, contracts, support tickets, images) tractable. A Big Data platform without an AI layer leaves most of this on the table.

Predictive maintenance and operational efficiency. Patterns that signal an impending failure are exactly the kind of weak, multi-variable signal that ML is good at and dashboards are bad at.

Personalization at scale. The combination of high-volume behavioral data and ML allows personalization to operate at the level of individuals rather than segments.

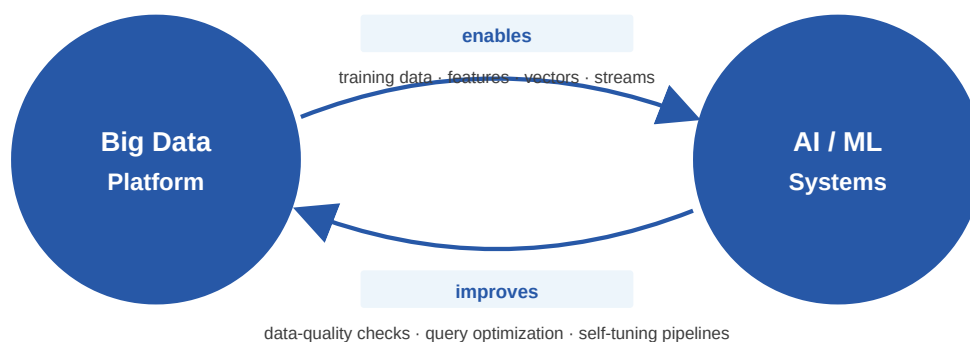


Figure 8. *The integration is bidirectional. Big Data makes AI possible; AI makes Big Data systems smarter.*

4.2 What data brings to AI

The integration is also bidirectional: AI improves Big Data systems (automated data quality checks, anomaly detection in pipelines, query optimization, intelligent caching), and Big Data systems make AI possible (training data at scale, feature stores, online serving). A team that views these as separate disciplines will under-invest in both.

5. Case study: Mercedes-AMG PETRONAS F1

Formula 1 is one of the most data-intensive industries in the world, and Mercedes-AMG PETRONAS makes a particularly clean case study because the team has publicly discussed its data architecture for nearly a decade.

5.1 The workload

Each Mercedes F1 car carries more than 400 sensors that together generate over 1.1 million data points per second, covering tire degradation, aerodynamic behavior, energy-recovery deployment, power-unit performance, and dozens of other variables. During a race weekend, this telemetry must be ingested in real time, processed both at the track and back at the factory, fused with simulation data and historical results, and turned into decisions that the pit wall can act on within seconds.

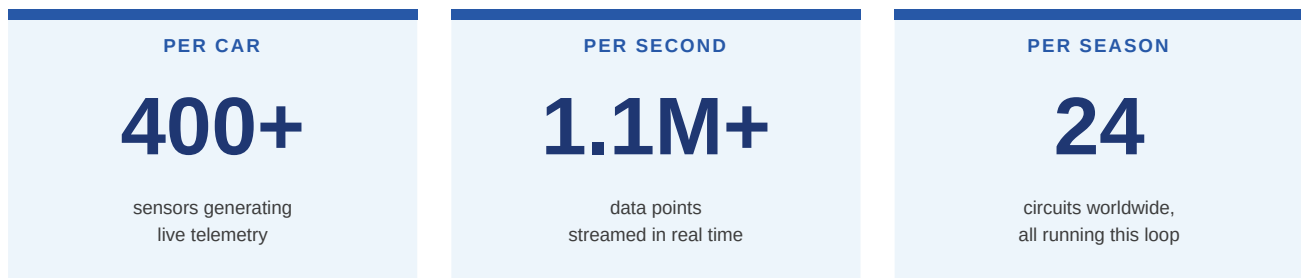


Figure 9. *The Mercedes-AMG PETRONAS F1 telemetry workload. Source: Microsoft (2026); Data Centre Magazine (2026).*

In other words, Mercedes runs essentially every workload in Section 2 simultaneously: high-velocity streaming, large-scale batch training, low-latency inference, and rich analytical querying.

5.2 Evolution of the platform

The team's data journey has unfolded in roughly three phases that mirror broader industry trends.

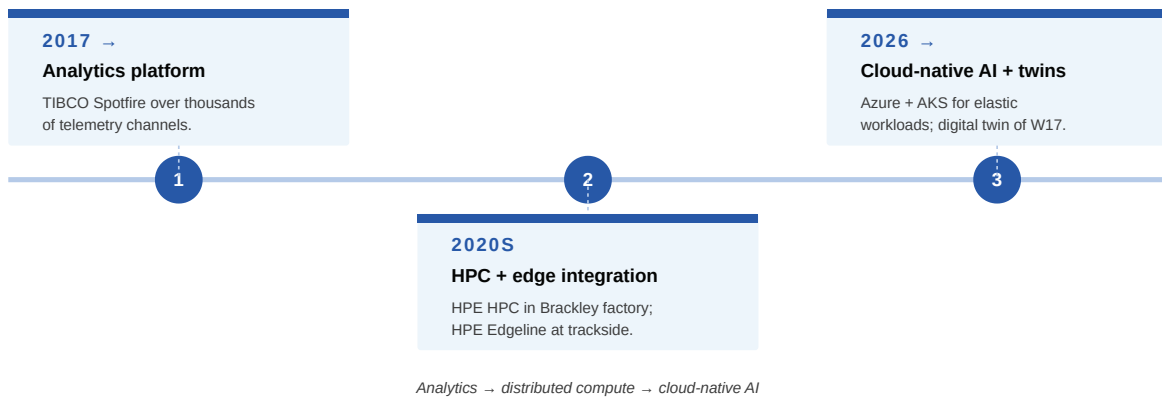


Figure 10. *The three phases of the Mercedes data and AI architecture, 2017 to 2026.*

Phase 1 — Analytics platform (from 2017). Mercedes partnered with TIBCO to build a unified analytics layer over its telemetry data, using TIBCO Spotfire to visualize thousands of telemetry channels and run race simulations. This phase was about turning raw telemetry into something humans could reason about.

Phase 2 — High-performance computing and edge integration (with HPE). As simulation workloads grew, Mercedes deployed HPE high-performance computing in its Brackley factory and HPE Edgeline converged systems trackside. The edge deployment matters because race-day decisions cannot wait for a round trip to a distant data center. The same architecture reduced full event simulations from three days to one.

Phase 3 — Cloud-native AI and digital twins (with Microsoft Azure, announced January 2026). Mercedes' most recent partnership places Microsoft Azure at the center of its operations for the 2026 F1 regulatory era. The team uses Azure Kubernetes Service to scale compute up during race weekends and down between them, and has piloted "intelligent virtual sensors" that use AI to estimate quantities the physical sensors cannot directly measure. In parallel, HPE infrastructure supports digital-twin simulation that completed a full virtual shakedown of the W17 car before the physical chassis was built.

5.3 The architecture in operation

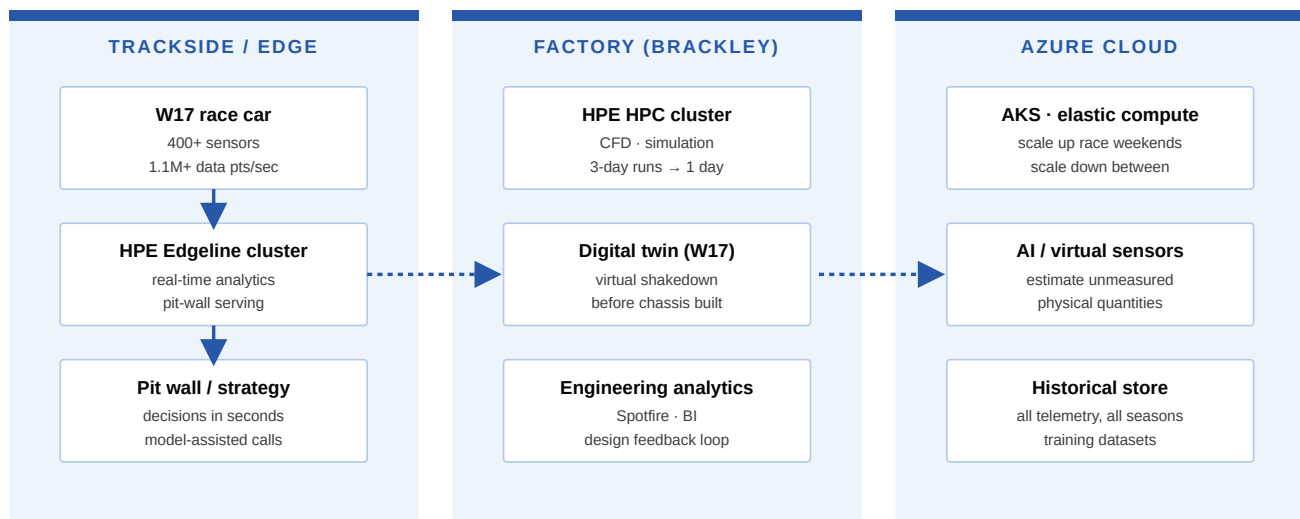


Figure 11. The Mercedes data + AI architecture, simplified. Three integrated tiers — trackside edge, factory HPC, and elastic cloud — each running the workload best suited to its latency, scale, and cost profile.

5.4 Mapping principles to the architecture

PRINCIPLE	HOW MERCEDES EXPRESSES IT
Separate storage / compute	Shared telemetry feeds simulation, analytics, AI training, and strategy — without duplication.
Edge + cloud continuum	HPE Edgeline trackside. HPE HPC in factory. Azure for elastic AI workloads.
Streaming + batch unified	Live race telemetry feeds strategy; the same data accumulates into next-race training sets.
Elasticity under constraint	F1 cost cap forces "do more with the same." AKS scales up only on race weekends.
AI integration	Virtual sensors, race-strategy models, full-car digital twin of the W17.
Sustainability	Net-zero IT targets; energy-aware workload placement across 24 circuits.

5.5 Lessons that translate beyond motorsport

Three lessons translate well beyond F1.

First, the architecture is the strategy. Mercedes does not treat data infrastructure as a back-office function; the choice of where to compute, how to store, and which workloads to push to the edge directly shapes what is

competitively possible on Sunday afternoon. The same is increasingly true in finance, healthcare, logistics, and manufacturing.

Second, AI is integrated, not bolted on. The virtual-sensor and digital-twin work only succeed because the underlying data platform is already in place. Organizations that try to "add AI" to a disorganized data foundation tend to discover that 80% of the work is foundational data engineering.

Third, constraints drive design. F1's cost cap, sustainability rules, and FIA regulations on simulation hours are not friction to be eliminated but structure to be designed around. Real architectures always operate under constraints — budget, latency, regulation, energy — and the discipline is to make those constraints visible early.

"Microsoft Azure and its AI capabilities will expand the team's existing high-performance computing and data capabilities, both at the factory and trackside, with scalable cloud and AI resources supporting simulation workloads, performance analysis, race strategy modeling and cross-team analytics." — Microsoft press release, January 2026

6. A framework for student exercises

For instructional use, the following questions can be applied to any organization a student is studying:

1. What are the dominant data velocities and volumes, and what does that imply for the storage and processing layer?
2. Where in the workflow is latency critical, and where is it not? What does that imply for edge vs. cloud placement?
3. Which decisions are made by humans, and which by models? What does that imply for the serving infrastructure and the human–machine interface?
4. What are the binding constraints (budget, regulation, sustainability, talent)? How does the architecture respect them?
5. What is the failure mode you most fear, and what does the platform do to detect or prevent it?
6. If a competitor adopts a fundamentally different architecture two years from now, how locked in is yours?

Students should be able to answer these for the Mercedes case after the reading, and then for an organization in their own domain.

7. Closing thought

CLOSING THOUGHT

Big Data infrastructure and AI infrastructure used to be discussed as separate disciplines. They are now **one discipline**, because the same engineering choices — about storage, compute, latency, governance, and cost — determine both what analytics you can run and what models you can train and serve.

A graduate course on the subject is therefore less about memorizing the current tool landscape (which will change) and more about internalizing the trade-offs that recur across every generation of the technology. The Mercedes case is useful precisely because it shows those trade-offs being made openly, under real competitive pressure, in a setting where the consequences of a wrong choice are visible to 800 million viewers.

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